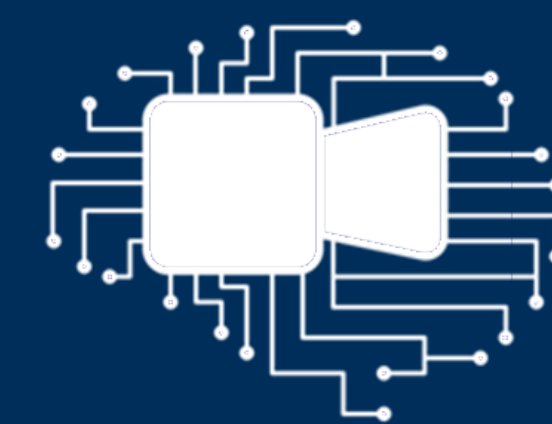




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The logo for BMVC 2025, featuring the text "BMVC" in a large, bold, white sans-serif font, with "2025" in a smaller, bold, white sans-serif font directly below it. The logo is set against a dark blue background.

Why Audio and Video Matter for Saliency:

Limits of Current Audio-Visual Models:

Computational Bottlenecks Block Deployment:

Need for a Balanced Solution:

- There is a clear gap for an AV saliency framework that (1) **captures long-range dependencies**, (2) **aligns audio–visual signals adaptively**, and (3) **delivers strong accuracy under tight compute budgets**.

Datasets: 6 AV (DIEM, Coutrot1/2, AVAD, ETMD, SumMe) + 2 visual (DHF1K, UCF-Sports);

Results: new SOTA - 24/24 AV wins; DTFSal-V #1 vs visual SOTAs;

Efficiency: lighter-weight than DiffSal yet better.

Table 1. Comparison with previous methods on six audio visual saliency datasets.

Method	#Parameters	DIEM				Control1				Control2			
		CC ↑	NSS ↑	AUC-J ↑	SIM ↑	CC ↑	NSS ↑	AUC-J ↑	SIM ↑	CC ↑	NSS ↑	AUC-J ↑	SIM ↑
STAVIS	20.76M	0.579	2.26	0.883	0.482	0.472	2.11	0.868	0.393	0.734	5.28	0.958	0.511
ViNet	33.97M	0.632	2.53	0.899	0.498	0.560	2.73	0.889	0.425	0.754	5.95	0.951	0.493
TSFP-Net	-	0.651	2.62	0.906	0.527	0.571	2.73	0.895	0.447	0.743	5.31	0.959	0.528
CASP-Net	51.62M	0.655	2.61	0.906	0.543	0.561	2.65	0.889	0.456	0.788	6.34	0.963	0.585
DiffSal	76.57M	0.660	2.65	0.907	0.543	0.638	3.20	0.901	0.515	0.835	6.61	0.964	0.625
MSPI	105.25M	0.671	2.71	0.905	0.545	0.605	2.98	0.899	0.473	0.799	6.40	0.964	0.5961
DAVS	-	0.580	2.28	0.884	0.483	0.481	2.19	0.869	0.400	0.734	4.984	0.960	0.512
DTFSal(V)	40.73M	0.677	2.76	0.910	0.548	0.623	3.21	0.904	0.505	0.842	6.42	0.965	0.618
DTFSal	49.08M	0.682 ^{+1.64%}	2.98 ^{+9.23%}	0.921 ^{+1.54%}	0.562 ^{+3.12%}	0.661 ^{+3.61%}	3.31 ^{+3.44%}	0.911 ^{+1.11%}	0.523 ^{+1.55%}	0.845 ^{+1.20%}	6.72 ^{+1.66%}	0.971 ^{+0.73%}	0.649 ^{+3.84%}
Method	#Parameters	AVAD				ETMD				SumMe			
		CC ↑	NSS ↑	AUC-J ↑	SIM ↑	CC ↑	NSS ↑	AUC-J ↑	SIM ↑	CC ↑	NSS ↑	AUC-J ↑	SIM ↑
STAVIS	20.76M	0.608	3.18	0.919	0.457	0.569	2.94	0.931	0.425	0.422	2.04	0.888	0.337
ViNet	33.97M	0.674	3.77	0.927	0.491	0.571	3.08	0.928	0.406	0.463	2.41	0.897	0.343
TSFP-Net	-	0.704	3.77	0.932	0.521	0.576	3.07	0.932	0.428	0.464	2.30	0.894	0.360
CASP-Net	51.62M	0.691	3.81	0.933	0.528	0.620	3.34	0.940	0.478	0.499	2.60	0.907	0.387
DiffSal	76.57M	0.738	4.22	0.935	0.571	0.652	3.66	0.943	0.506	0.572	3.14	0.921	0.447
MSPI	105.25M	0.702	3.92	0.936	0.533	0.623	3.35	0.940	0.471	0.502	2.61	0.909	0.380
DAVS	-	0.610	3.19	0.919	0.458	0.600	2.96	0.932	0.425	0.422	2.28	0.888	0.338
DTFSal(V)	40.73M	0.742	4.21	0.938	0.559	0.653	3.57	0.939	0.499	0.561	2.91	0.911	0.441
DTFSal	49.08M	0.746 ^{+1.09%}	4.36 ^{+3.32%}	0.939 ^{+0.33%}	0.595 ^{+4.07%}	0.667 ^{+2.30%}	3.82 ^{+4.37%}	0.937 ^{+1.48%}	0.531 ^{+4.94%}	0.573 ^{+0.175%}	3.20 ^{+2.91%}	0.927 ^{+0.65%}	0.449 ^{+0.49%}

- **AV saliency:** multiscale visual + audio.
- **Encoders:** MViT2; log-Mel $\rightarrow$ VGGish + temporal Transformer.
- **Refinement:** top-down + lateral 3D convs.
- **Fusion/decoding:** AMFB (local/global/deformable) + tri-stream score; hierarchical decoders.
- **Tokens:** LTEB gates salient; DLTFB shifts/fuses for long-range.

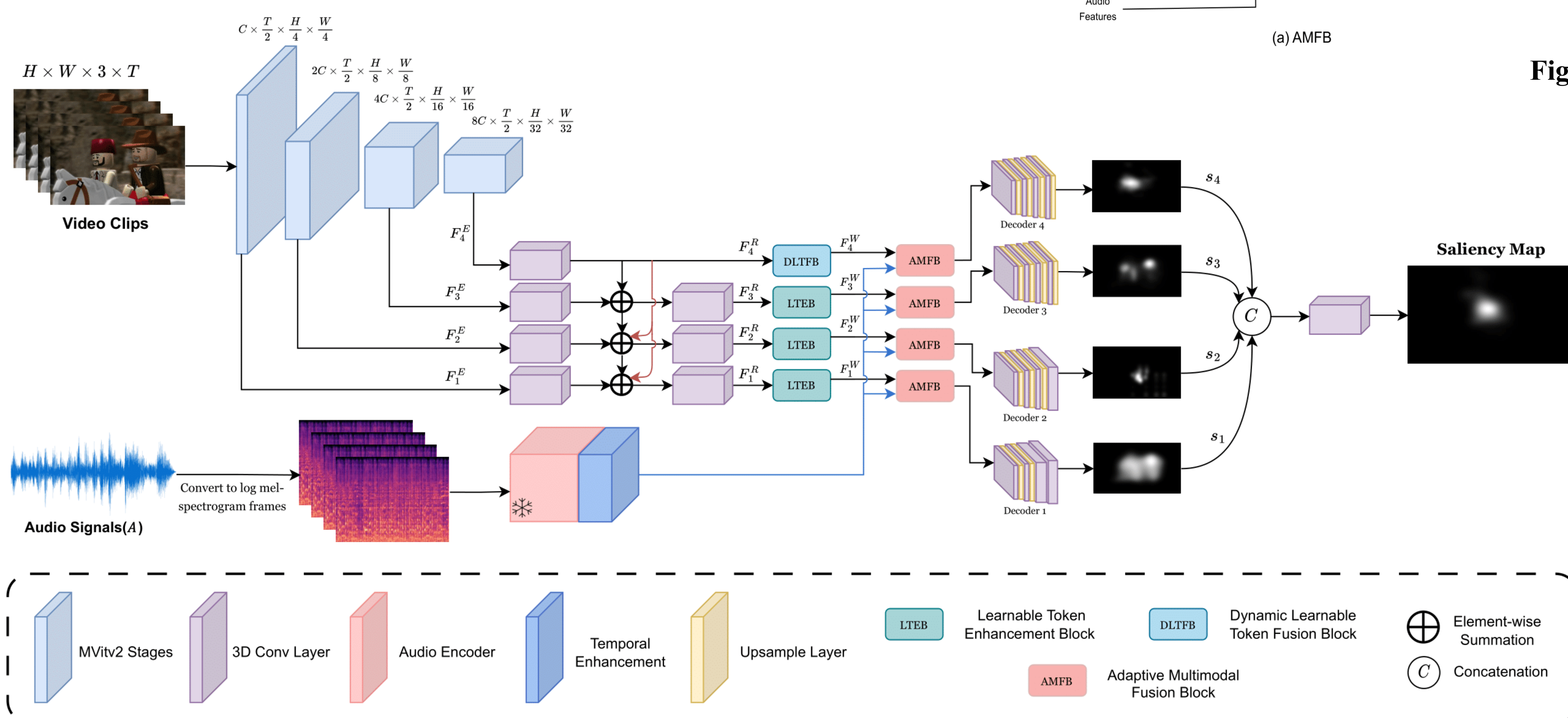


Figure 1. Overview of our SUM model.

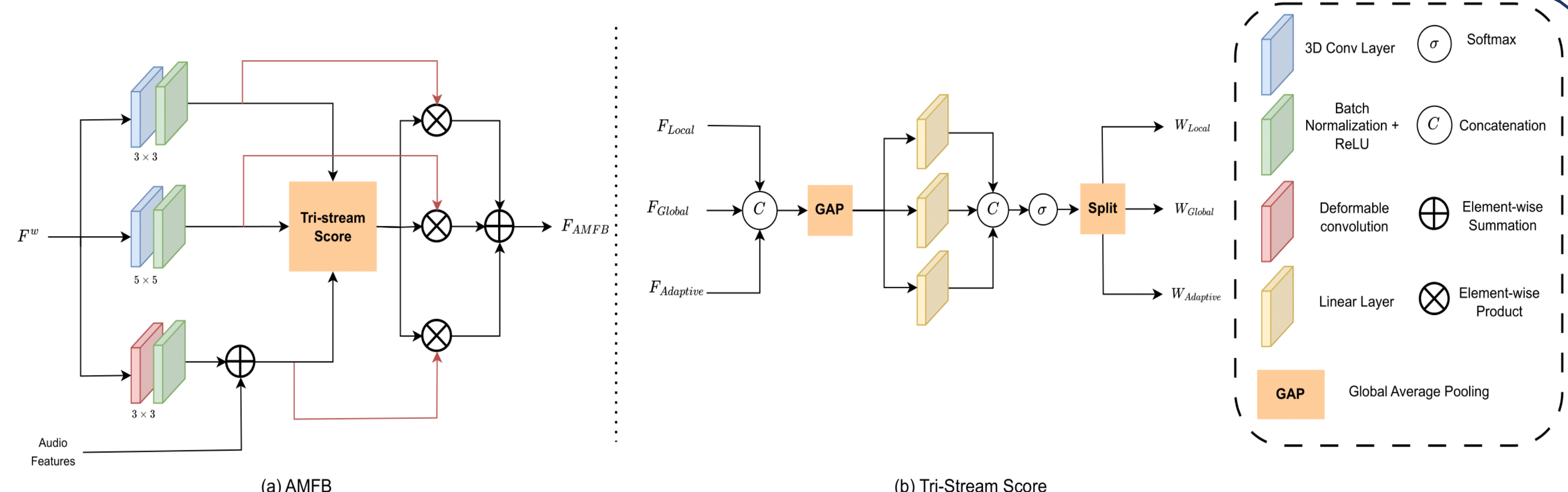


Figure 2. Overview of AMFB.

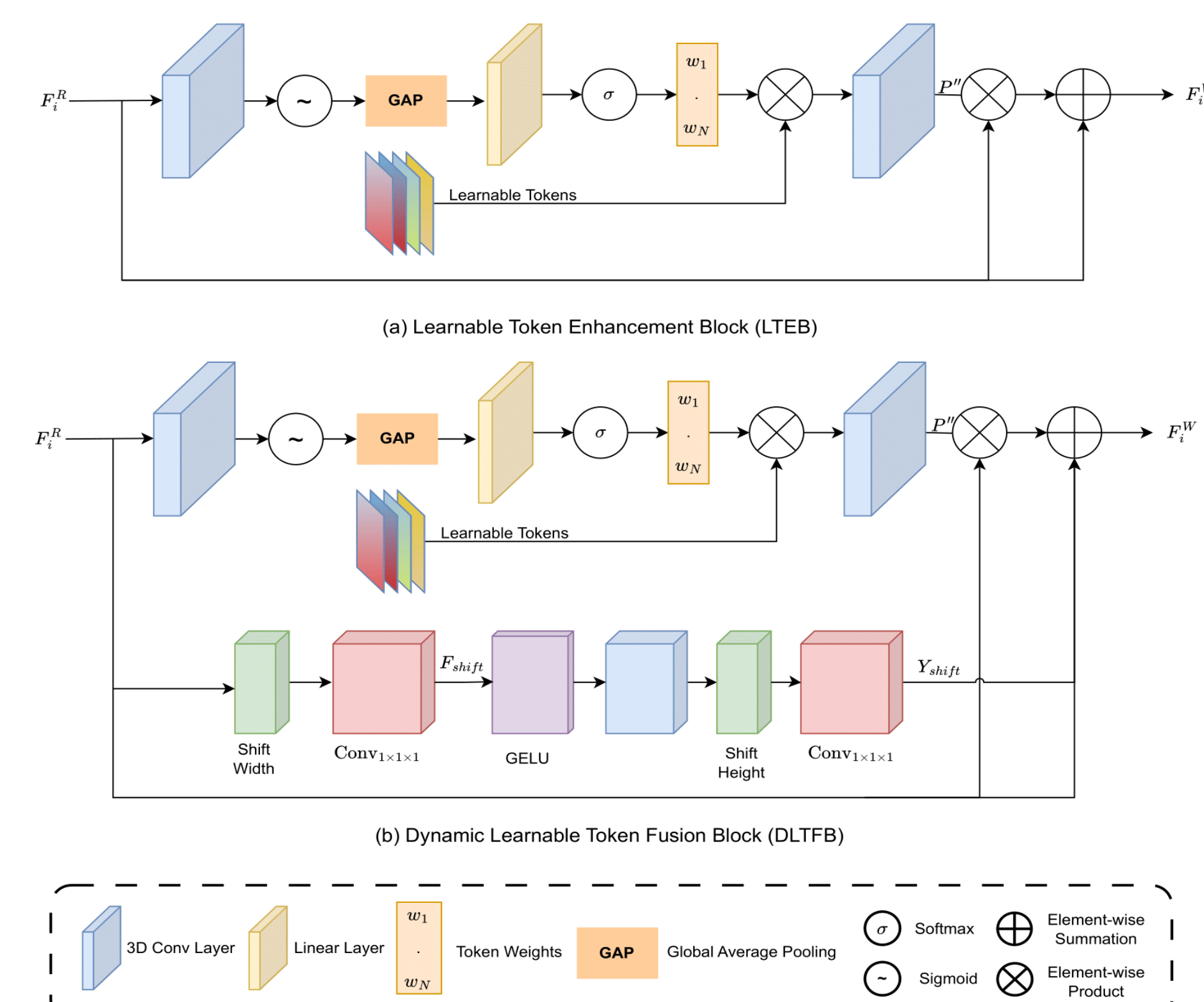


Figure 3. Overview of LTEB and DLTFB.

Table 2. Visual-only baseline vs. visual SOTAs on DHF1K.

Model	DHF1K			
	CC $\uparrow$	NSS $\uparrow$	SIM $\uparrow$	AUC-J $\uparrow$
DeepVS	0.344	1.911	0.256	0.856
ACLNet	0.434	2.354	0.315	0.890
STRA-Net	0.458	2.558	0.355	0.895
TASED-Net	0.470	2.667	0.361	0.895
SalSAC	0.479	2.673	0.357	0.896
UNISAL	0.490	2.776	0.390	0.901
ECANet	0.500	2.814	0.385	0.903
HD2S	0.503	2.812	0.406	0.908
ViNet	0.511	2.872	0.381	0.908
TSFP-Net	0.517	2.967	0.392	0.912
GFNet	0.526	2.995	0.379	0.913
STSANet	0.529	3.010	0.383	0.913
TMFI-Net	0.546	3.146	0.407	0.9153
TinyHD-S	0.492	2.873	0.388	0.907
THTD-Net	0.547	3.138	0.406	0.9152
TempVST	0.494	3.73	0.373	0.904
ViNet-E	0.549	3.13	0.409	0.922
DiffSal(V)	0.533	3.066	0.405	0.918
HSFI-Net	0.548	3.163	0.411	0.914
DTPSal(V)	0.561$\pm$0.37%	3.205$\pm$1.33%	0.436$\pm$6.08%	0.923$\pm$0.55%

Table 3. FLOPs, input size, and model size vs prior visual & AV saliency methods.

Model	Input Size	FLOPs (G)	Model Size (MB)
Visual Models			
DeepVS	448×448	819.5	344
STRA-Net	256×224	55	324
TASED-Net	256×224	91.5	82
UNISAL	288×224	14.4	146
HD2S	192×128	22	155
ViNet	192×128	115	152
STSA-Net	384×224	493	245
TMFI-Net	384×224	358	234
THTD-Net	384×224	-	220
TM2SP-Net	384×224	1330	700
ViNet-E	384×224	-	183.84
HSFI-Net	384×224	-	253
DTFSal(V)	384×224	259.57	155.37
Audio-Visual Models			
ACLNet	224×224	164	250
TSFP-Net	352×192	316	192
STAVIS	384×224	15.31	79.19
Casp-Net	384×224	283.35	196.91
DiffSal	384×224	834	351
DTFSal	384×224	297.91	187.26

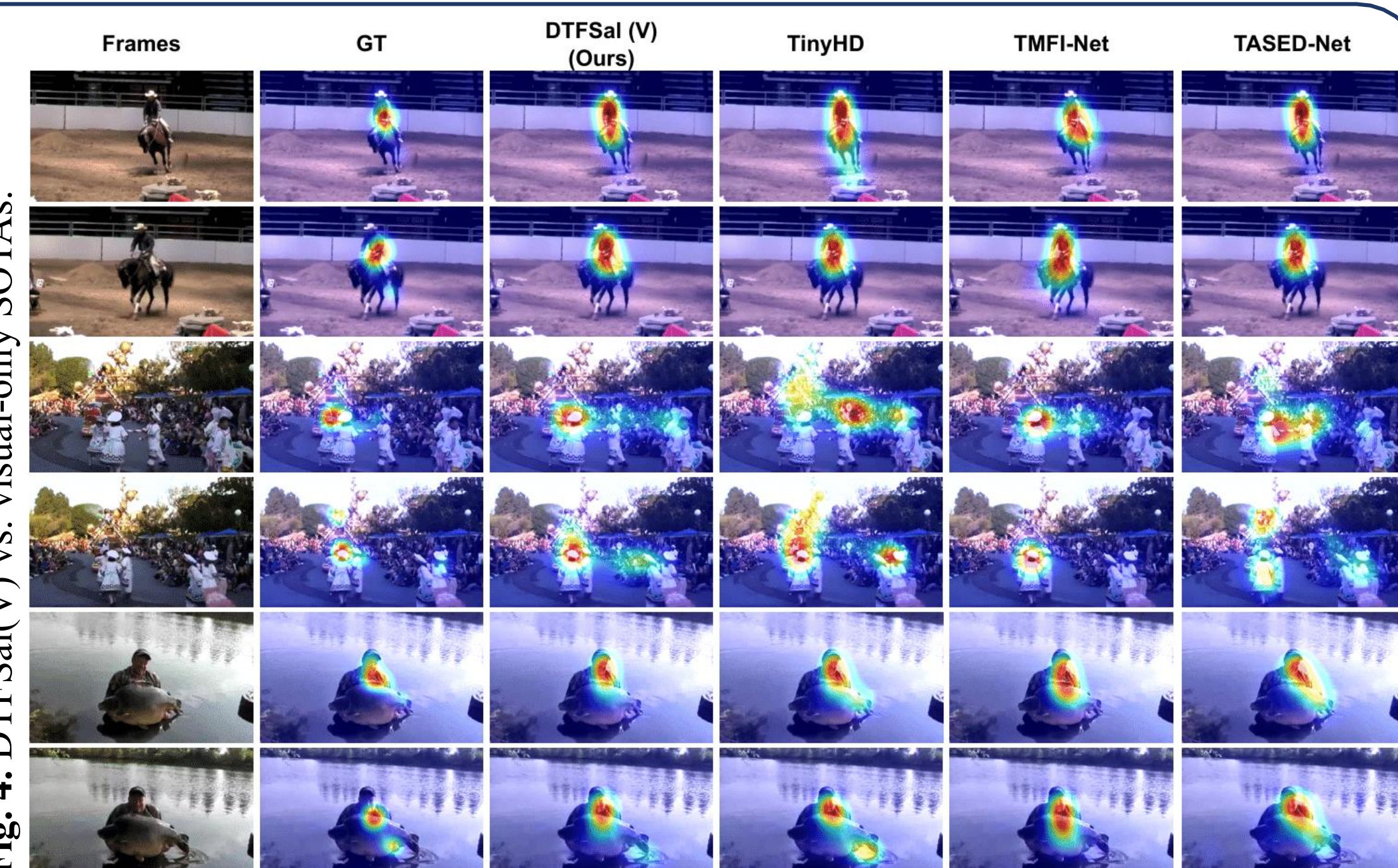


Fig. 4. DTFSal(V) vs. visual-only SOTAs.

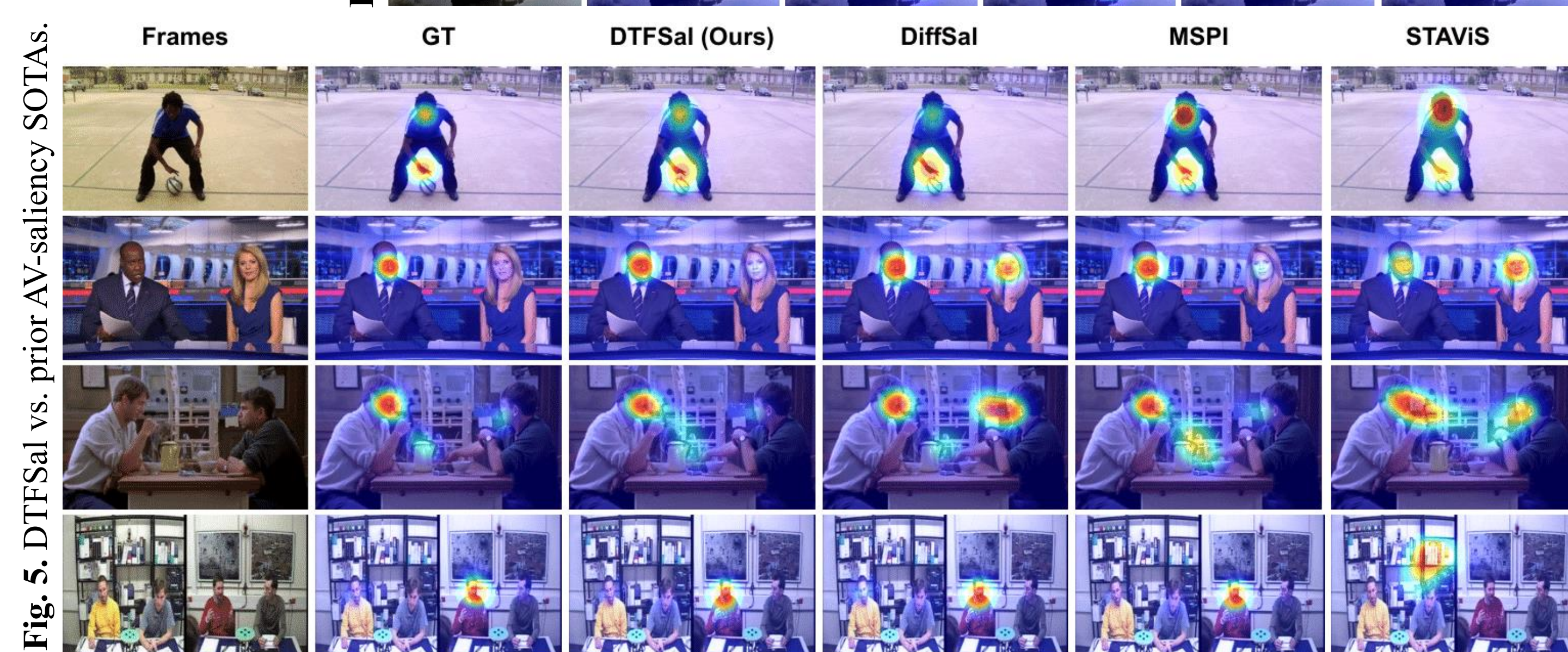


Fig. 5. DTFSal vs. prior AV-saliency SOTAs.

For more Information:



SCAN ME